

Qingyun Yang

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🌐 <https://anne-yang.github.io/qingyun-yang.github.io/>

A senior undergraduate researcher interested in 3D computer vision, medical image analysis, and AI in healthcare.

EDUCATION

Vanderbilt University B.S.

Aug 2022 - Present (expected May 2026)

Majors: Computer Science, Mathematics | *Minors:* Data Science

Nashville, Tennessee

GPA: 3.948/4 (Dean's List All Semesters)

Honors: MICCAI 2025 NIH Registration Grant, CRA Outstanding Undergraduate Researchers (Honorable Mention), VISE Fellowship 2024 (\$8000 funding), VISE Fellowship 2025 (\$8000 funding)

PUBLICATION

Q. Yang*, F. Li*, J. Xu, Z. Liu, S. Sridhar, W. Jin, J. Du, J.S. Heiselman, M.I. Miga, M. Topf, J.Y. Wu. "Augmented Reality-based Guidance with Deformable Registration in Head and Neck Tumor Resection." in International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2025). (oral presentation)

Q. Yang, A. Acar, J. M. Ringel, J.S. Heiselman, M.I. Miga, M. Topf, J.Y. Wu. "Nonrigid Alignment of En Bloc Tissue Specimen to Resection Bed to Enhance Correspondence for Re-Resection Guidance." SPIE Medical Imaging, 2025. (oral presentation)

H. Li, J. Wang, N. Kumar, T.E. Ertop, J. d'Almeida, D. Lu, A. Acar, J. Han, **Q. Yang**, J.Y. Wu, R.J. Webster III, I. Oguz. "Automated Segmentation of Central Airway Obstruction from Endoscopic Video Stream with Deep Learning." in SPIE Medical Imaging, 2025.

D. Su, **Q. Yang**, K.D. Van Schaik, L.W. Remedios, Y. Zhang, T. Li, F. Maldonado, K.L. Sandler, B.M. Dawant, B.A. Landman. "BPPR: A Framework for Content Navigation in Multi-Contrast Body CT Images using Deep Regression Models." in IEEE Transactions on Biomedical Engineering. (under review)

M. Ringel, A. Acar, **Q. Yang**, M. Aweeda, C. Fassler, J. Heiselman, J.Y. Wu, M. Topf, M. Miga. "Specimen-to-tumor bed deformable registration to inform re-resection in otolaryngologic procedures." in Medical Image Understanding and Analysis, vol. 92, Frontiers Media SA, Manchester, UK, 2024.

RESEARCH EXPERIENCE

3D Guidance via Nonrigid Registration for Accurate Tumor Re-Resection

January 2024 – Present

Independent Researcher | Computer Vision, Image Guidance Surgery

Vanderbilt University

- Led an independent research project under **Prof. Jie Ying Wu**, developing a surgical guidance framework for tumor specimen intraoperative deformation correction, which was ultimately deployed on Augmented Reality HoloLens to enhance intraoperative visualization
- Reconstructed 3D specimen meshes using nonrigid registration with regularized Kelvinlets, achieving a **54%** improvement in alignment
- Developed a novel registration pipeline that utilizes both pre- and post-resection site surfaces as targets, improving accuracy by up to **33%** over previous methods
- Conducted **8** cadaver studies, collecting **23** En Bloc tissue specimens and reconstructing corrected 3D specimen models. Contributed to more accurate post-operative analysis of tissue deformation

Content Navigation in Body CT Images using Deep Regression Models

August 2024 – January 2025

Research Assistant | Machine Learning, Medical Imaging

Vanderbilt University

- Worked under the mentorship of **Prof. Bennett A. Landman** to analyze the generalizability of Body Part Regression (BPR) technology across multi-contrast CT data.
- Pretrained and fine-tuned deep regression models on multi-contrast datasets, rigorously evaluating the influence of Global-Intensity-Nonlinear augmentation on model generalizability.

Progression-Aware Data Generation for Multi-Modality Brain Imaging

April 2025 – Present

Independent Researcher | Generative Model, Medical Imaging

Johns Hopkins University (Remote)

- Proposed a generative framework that disentangles anatomical variance and pathology progression to enable tumor growth modeling.
- Currently building the core framework and preparing for model training and validation using multi-modality brain imaging data.

Video and Depth Data Collection for Central Airway Obstruction

May 2024 - August 2024

Research Assistant

Vanderbilt University

- Developed a data collection pipeline for recording with the da Vinci robot using ROS and rosbag, streamlining data acquisition processes
- Conducted data collection using the Polaris Spectra tracker and Spectrum HOPKINS® Telescope attached to a Sentech (Omron) STCHD4K camera

Technical Skills

Languages: Python, C++, Java, R, Racket, Prolog, MATLAB Relavent

Courses: Foundations of Machine Learning, Image Processing, Advanced Linear Algebra, Probability and Statistics for Engineering, Multi-variable Calculus, Methods of Ordinary Differential Equations, Intermediate Software Design, Data Structures, Algorithm, Computer Architecture, Electricity and Magnetism, Mechanics

ADDITIONAL EXPERIENCE

Vanderbilt Computer Science department

August 2023 - Present

Teaching Assistant

Vanderbilt University

- Conducted 4 hours of weekly office hours, addressing inquiries from over 200 students across 3 sessions, enhancing the students’ understanding of programming concepts
- Evaluated assignments and exams, offering detailed feedback on grammar, logic, and coding style, ensuring students develop strong programming skills and promoting best practices in code quality.